

Document fields detection with computer vision

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Abstract—In this paper, we present a computer vision approach for the processing and enhancement of document images. The main goal is to improve the readability of digitized or photographed documents, ensuring that they can be accurately read and processed in the future.

Keywords—Computer vision, image treatment, digitalized documents.

I. INTRODUCTION

With the growth of digitized documents, it is essential to ensure that the captured images maintain the necessary readability for future processing. However, issues such as low-quality camera images, poor lighting, and the presence of noise can compromise the clarity of the information in documents. In this context, computer vision emerges as a powerful tool for image processing and enhancement. Through preprocessing techniques such as the use of filters and edge detection algorithms, it is possible to improve the visual quality of documents, making them easier to read and store.

II. SYSTEM DESCRIPTION

The developed system aims to analyze and process images using computer vision techniques. The executed steps follow a structured flow, starting with the import of necessary libraries, such as OpenCV for image processing, NumPy for matrix operations, and Matplotlib for visualization of results.

Next, image reading is performed, which is initially loaded in color and subsequently converted to grayscale, facilitating processing in the following steps. In the pre-processing stage, a mean and median filters are applied to smooth the image and reduce noise, promoting a cleaner base for subsequent analyses.

In the edge detection phase, different algorithms are tested, with Roberts appearing to be the most appropriate due to its ability to identify contours with good precision and low computational cost. This edge detector is capable of highlighting abrupt intensity transitions in the image, which usually correspond to boundaries between different regions of interest. Soon after, segmentation takes place, where the image is binarized through thresholding and then morphological operations, such as closing, are applied to correct flaws and join nearby regions. This flow allows for precise visual and quantitative analysis of the elements present in the image, demonstrating the efficiency of the techniques used in the system.

The first step is to convert the image to grayscale, which makes it easier to apply filters and extract the text.

After applying the mean and median filters, which help smooth the image for better text recognition, we can now apply a threshold filter to convert the image into a binary format.



Fig. 1
ORIGINAL IMAGE



Fig. 2
IMAGES IN GRAYSCALE, BLURED USING 3X3 KERNEL AND MEDIAN FILTER USING KERNEL 3x3

This aids in recognizing the letters in the text. Threshold values ranging from 120 to 200 were tested, where pixel values below the threshold become black and those above become white. Among the tested values, a threshold of 155 provided the best result, offering a good balance between accurate letter recognition and the removal of non-relevant information.

We can try applying Roberts and Canny filters for edge detection with different values for the threshold1 and threshold2 parameters for the Canny, but the results tend to make the image less smooth for text detection and often introduce additional noise on Canny while the Roberts doesn't add the

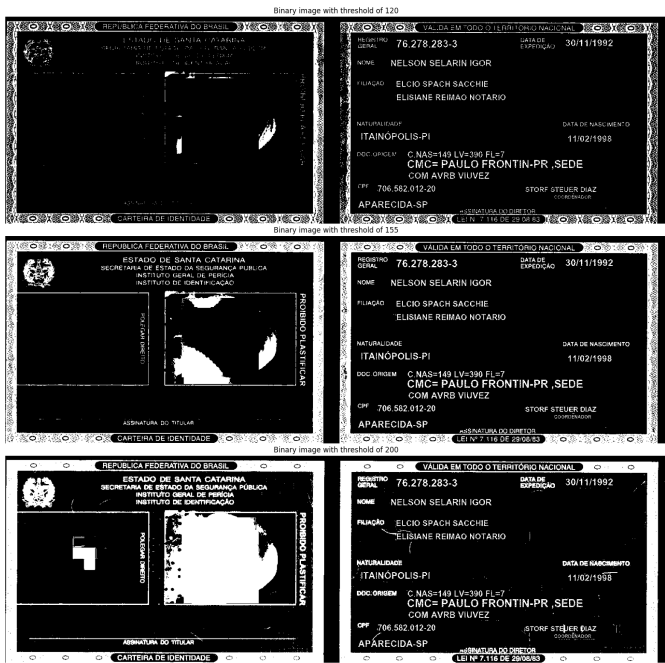


Fig. 3

BINARY IMAGES WITH THRESHOLD VALUES OF 120, 155 AND 200



Fig. 4

CANNY FILTERS WITH DIFFERENT THRESHOLD1 AND THRESHOLD2 VALUES.

noise making it a lot better than Canny for this use.

After applying the threshold filter to convert the image to binary, we can extract a specific region and use a dilation morphological filter to enhance the size of the target objects—in this case, the letters. This helps improve the text recognition process. Without dilation, the letters often contain small black spots inside them. After the first dilation, these spots are noticeably reduced, and with a second dilation, the letter edges become more defined, making them easier to recognize.

With these processes, we obtain a much clearer and smoother image, which facilitates the segmentation and recognition of the text in each field of the document.

III. PROPOSED METHOD

To improve the quality of documents and ensure their legibility, this work proposes the use of computer vision techniques focused on image preprocessing. Among the techniques used are smoothing filters, edge detection algorithms, and morphological operations. These techniques allow for noise reduction, enhancement of relevant regions, and correction of imperfections that hinder readability. The goal is to improve the clarity of the image content, facilitating reading for the use of automatic recognition systems.

IV. EXPECTED RESULTS

It is expected that the application of image processing techniques proposed in this work will result in a significant improvement in the legibility of scanned or photographed documents. With the use of filters, an efficient reduction of visual noise is expected. Edge detection should allow better

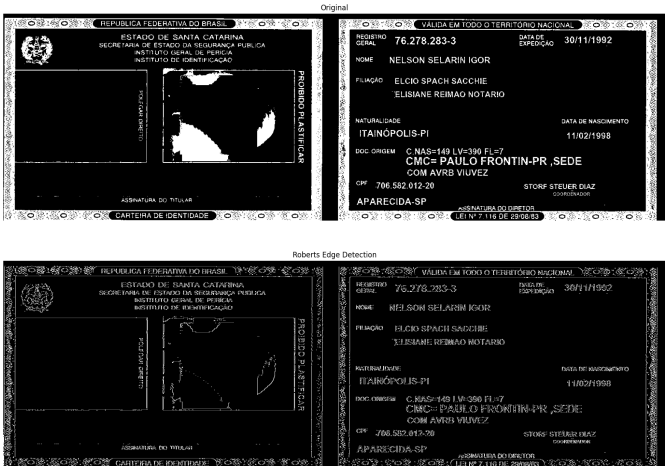


Fig. 5

ORIGINAL IMAGE AND IMAGE AFTER APPLYING ROBERTS FILTER



Fig. 6

IMAGE SEGMENTS WITHOUT DILATION, WITH 1 DILATION AND WITH 2 DILATIONS

highlighting of regions of interest, especially textual characters. Morphological operations should contribute to image refinement, correcting flaws, and removing small imperfections. As a final result, the processed images should present greater contrast between text and background, more defined lines, and absence of unwanted elements, enabling clearer reading by users and recognition systems.

V. CONCLUSIONS

This project made it possible to apply, in practice, fundamental techniques studied in the Computer Vision course for processing document images, such as the use of filters, edge detection, and morphological operations, reinforcing the importance of image pre-processing in applications like digitization. It also provided a clear understanding of how simple methods can yield significant results in solving real-world problems.

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